Rate of adverse cardiovascular events in breast cancer patients receiving chemotherapy and targeted therapy

Shuang Yang, MS[1], Dongyu Zhang PhD[2], Jiang Bian, PhD[1], Dejana Braithwaite, PhD[1], Yi Guo, PhD[1]
[1]University of Florida, Gainesville, Florida, USA; [2]Johnson & Johnson, New Brunswick, NJ, USA

Background

- Breast cancer is the most common cancer and the second leading cause of cancer death in women in the United States
- Prior studies have shown that use of adjuvant therapy significantly increases breast cancer cure rate and reduces recurrences
- However, adjuvant therapy, especially chemotherapy and targeted therapy, may cause cardiotoxicity, one of the most important adverse reactions of adjuvant therapy
- Patients with a higher burden of comorbidities may have a higher cardiotoxicity risk since they have poorer prognosis compared to their healthier counterparts
- The current study aimed to assess how risk of cardiotoxicity in breast cancer patients on chemotherapy and targeted therapy varied by burden of comorbidities

Methods

Study Cohort

- In electronic health records (EHRs) in the OneFlorida+ Clinical Research Network, we identified 2,198 breast cancer patients who received chemotherapy and target therapy based on HCPCS, RXCUI and NDC codes

Outcomes

- Cardiovascular events were identified from the first date of chemo/targeted therapy to 3 months after the last date of chemo/targeted therapy using ICD codes
- The first date of chemo/targeted therapy was defined as the date of the first records of chemo/targeted therapy, the last date of chemo/targeted therapy is defined as the date of the last code of chemo/targeted therapy and after 90 days of the date there are no chemo/targeted therapy code occur

Statistics analysis

- We used 12 months data before first chemo/targeted therapy date to calculate Charlson comorbidity index (CCI) scores to reflect the pre-existing comorbidities status of patients
- We calculated cardiovascular event rates by CCI and built multivariable logistic models to examine the association of multiple factors with cardiotoxicity, including CCI, age, race-ethnicity, body mass index (BMI), outpatient visits, tumor grade, stage, size, at tumor diagnosis, and census tract-level poverty and residence

Results

- In the 2,198 patients, Overall, 9.5% (N=208) of the patients had any cardiotoxicity event after chemo/targeted therapy.
- The cardiotoxicity rates by CCI were 9.1% (CCI=0), 9.4% (CCI=1), and 10.5% (CCI=1+).
- Although statistically non-significant, patients with higher CCI were more likely to experience cardiotoxicity events (For CCI = 1 vs. 0: odds ratio [OR] = 1.20, 95% confidence interval [CI] = 0.85-1.71; For CCI = 1+ vs. 0: OR = 1.34, 95% CI = 0.87-2.08).
- Non-Hispanic black (OR=1.77; p=0.005), diagnosed with distant stage (OR=2.14; p=0.001) were more likely to have cardiotoxicity event compared to non-Hispanic white and diagnosed with localized.

Table 2: Regression estimating association between CCI cardiotoxicity event

	breast cancer patients with chemo/targeted therapy (N=2,138)	
Variables	**Adjusted OR (95% CI)**	***p*-value**
CCI[1]		
1 vs. 0	1.20 (0.85, 1.71)	= 0.299
2+ vs. 0	1.34 (0.87, 2.08)	= 0.187
Age at diagnosis		
60-69 vs. <60	0.74 (0.53, 1.03)	= 0.077
>70 vs. <60	0.69 (0.46, 1.03)	= 0.071
Race-ethnicity[2]		
NHB vs. NHW	1.77 (1.28, 2.44)	= 0.005
NHO vs. NHW	0.66 (0.28, 2.44)	= 0.336
Hispanics vs. NHW	0.59 (0.28, 1.54)	= 0.093
Stage		
Regional vs Localized	1.31 (0.94, 1.82)	= 0.113
Distant vs Localized	2.14 (1.39, 3.28)	= 0.001

Table 1: Respondents' characteristics.

	Overall (N=2,198)	No cardiotoxicity Event (N=1,990, %=90.5)	Has cardiotoxicity Event (N=208, %=9.5)	P-value*
CCI group1				
0	954 (43.84)	867 (90.9)	87 (9.1)	0.726
1	874 (39.8)	792 (90.6)	82 (9.4)	
2+	370 (16.08)	331 (89.5)	39 (10.5)	
Age at diagnosis				
<60	914 (41.6)	811 (88.7)	103 (11.3)	0.046
60-69	805 (36.6)	737 (91.6)	68 (8.5)	
>70	479 (21.8)	442 (92.3)	37 (7.7)	
Race-Ethnicity2				
NHW	1339 (60.9)	1223 (91.3)	116 (8.7)	<0.001
NHB	504 (22.9)	431 (85.5)	73 (14.5)	
Hispanic	220 (10.0)	208 (94.6)	12 (5.5)	
NHO	103 (4.7)	97 (94.2)	6 (5.8)	
Stage				
Localized	965 (43.29)	892 (92.4)	73 (7.6)	0.004
Regional	860 (39.1)	775 (90.1)	85 (9.9)	
Distant	345 (15.7)	298 (86.4)	47 (13.6)	
Grade				
Grade I	141 (6.4)	132 (93.46)	79 (6.4)	0.447
Grade II	505 (23.0)	452 (89.5)	53 (10.5)	
Grade III	712 (32.4)	638 (89.6)	74 (10.4)	
Other[3]	259 (11.8)	236 (91.1)	23 (8.9)	
Size				
0mm-10mm	200 (9.1)	186 (93.0)	14 (7.0)	0.146
>10mm-30mm	732 (33.3)	661 (90.3)	71 (9.7)	
>30mm	582 (26.5)	514 (88.3)	68 (11.7)	
Poverty				
<10%	598 (27.2)	547 (91.5)	51 (8.5)	0.532
10%-20%	845 (38.4)	758 (89.7)	87 (10.3)	
>=20%	435 (19.8)	392 (90.1)	43 (9.9)	
RUCA				
Rural	536 (24.4)	475 (88.6)	61 (11.4)	0.106
Urban	1342 (61.1)	1222 (91.1)	120 (8.9)	
BMI				
<25	350 (15.9)	331 (94.6)	19 (5.4)	0.100
25-29.9	514 (23.4)	468 (91.0)	46 (8.9)	
>=30	663 (30.2)	603 (91.0)	60 (9.1)	
Outpatient visit	16.7 (20.6)	16.5 (20.3)	18.0 (20.6)	0.343

Conclusion

Although statistically non-significant, the rate of cardiotoxicity event in individuals with higher burden of comorbidity was slightly higher than that in the low burden of comorbidity individual.

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